

Unit-II

RDBMS — Relational database management system. RDBMS is a program that stores & manage data in structured tables.

Data is organized into rows (records) & column (attributes) & relationship are established between tables using keys.

It is a software program used to store, manage, & retrieve data.

Some of the most well-known RDBMSs include MySQL, PostgreSQL, MariaDB, Microsoft SQL Server, & Oracle Database.

Domain — Permitted set of values.

In a DBMS, a domain defines the set of allowable values an attribute (column) can legally contain.

Key Components of a Domain

- Data Type :- Specifies the fundamental type of data, such as INTEGER, VARCHAR, BOOLEAN or DATE.

Summary :

- **Format & length**: Restricts the size & structure can only hold a string of up to 50 characters. (e.g., a VARCHAR(50))
- **Range & Rules**: Limits values to a specific subset (e.g., an Age column might restrict values to integers between 18 and 65).

Examples of Domains 1.) Gender - Domain could be restricted to M, F or O.

2.) Percentage - Domain can only hold decimal or integer values between 0.0 & 100.0.

* **Degree of Relation** :- Total no. of attributes (Columns) $\Rightarrow$ A/c to below Table $\Rightarrow 6$

* **Cardinality of Relation** :- Total no. of tuples (rows) $\Rightarrow$ A/c to below Table $\Rightarrow 4$
No. of attributes
No. of tuple

Summary :



- In RDBMS, data is stored in multiple tables, which are linked through some common columns or fields.

Relational Algebra

Relational algebra is the theoretical foundation of database query languages like SQL. It consists of procedural operation that take one or two relations (tables) as input & output a new relation.

- Relational algebra is a query language used in DBMS to perform operations on tables (relations).

It tells the database:

- What data to find
- How to combine tables
- How to filter data.

It is the mathematical foundation of SQL.

- Relational algebra works on tables & produces a new table as output.

Input $\rightarrow$ Table

Output $\rightarrow$ New Table

Summary :

Six Fundamental Operation

1.) Select Operation (σ) — Used to select specific rows based on condition.

Ex:- Students having marks > 80
 $\sigma_{\text{marks} > 80}(\text{Student})$

Result:-

Roll No.	Name	Marks
1	Rahul	85
2	Priya	90

$\therefore$ Select only required rows.

2.) Product Operation (π) — Used to select specific columns.

Ex:- Show only Name column

$\pi_{\text{Name}}(\text{Student})$

Result:-

Name
Rahul
Priya
Aman

$\therefore$ Select only required columns.

Summary :



3.) Union Operation ($\cup$) — Combines rows of two tables.

Ex:- Student table $\cup$ Old Student table.

$\therefore$ Merge two tables.

4.) Set Difference ($-$) — Finds rows present in first table but not in second.

Ex:- Current Students — Passed Students

$\therefore$ Removes common rows.

5.) Cartesian Product ($\times$) — Combines every row of first table with every row of second table.

$\therefore$ All possible combinations.

6.) Rename Operation ($\rho(\text{table})$) — Rename operation is used to change the name of:

- Relation (table)
- Attributes (columns)

Ex:- Student

Roll No.	Name
1.	Rahul

Summary :

Now we rename student table as S .

S (Student)

$\therefore$ Student table ka new name = S .

Relational Algebra Queries

Relational Algebra is a procedural query language used in DBMS to retrieve & manipulate data from relational databases. It forms the theoretical foundation of SQL.

Difference between Relational Algebra & Relational Calculus.

Relational Algebra	Relational Calculus
• It is a procedural query language.	• It is a non-procedural query language.
• It tells how to retrieve data.	• It tells what data is required.
• Uses operation like Selection (σ), Projection (π), Join ($\bowtie$) etc.	• Uses logical formulas & conditions.
• User must specify the sequence of operations.	• User only specifies the desired result.
• Easier to understand & implement.	• More mathematical & logic-based.
• Forms the theoretical basis of SQL Processing.	• Forms the theoretical basis of SQL query specification.

Summary :



E-R Model Basic Concepts

The E-R Model is a high level data model used to design databases. It represents data in the form of entities, attributes & relationships.

1.) Entity :- An entity is a real-world object, person, place or thing about which data is stored.

Ex:- Teacher
Student
Course
Dept.

2.) Attribute :- An attribute is a property or characteristics of an entity.

Ex:- Student $\longrightarrow$ Rollno., Name, Age

Teacher $\longrightarrow$ Teacher ID, Name, Salary.

Types of Attributes

1.) Simple Attribute - Cannot be divided further (Age).

Summary :

- ii.) Composite Attribute — Can be divided into smaller parts.
(Name → First Name, Last- Name)
- iii.) Single-valued Attribute — Has ~~multiple~~ ^{one} value.
(~~Phone No.~~) (Roll no.)
- iv.) Multi-valued Attribute — Has multiple value. (Phone No.)
- v.) Derived Attribute — Calculated from another attribute
(Age from Date of birth).
- 3.) Relationship : — A relationship shows the association between two or more entities.
Ex:- Teacher teaches Course.
- 4.) Entity Set :- A collection of similar entities.
Ex:- All students in a college form a student entity set.
- 5.) Relationship Set :- A collection of similar relationships.
Ex:- All "enrolls in" relationship between students & courses.

Summary :

6.) Cardinality Ratio :- It specifies how many entities can participate in a relationship.

Types :- (1:1, M:1, 1:M, M:N)

Design Process

The database design process is the step by step method used to create an efficient & organized database.

Steps :- 1.) Requirement Analysis

2.) Conceptual Design

3.) Logical Design

4.) Normalization

5.) Physical design

6.) Implementation & Testing

7.) Maintenance.

Memory Trick :- RCLNPIM

→ Rohan Cycle Le Nikla, Phin Ice-cream Mila.

Summary :

Constraints

Constraints are the rules applied to a database table to ensure that the data stored is accurate, valid & consistent.

Types:- 1.) Domain Constraints

2.) Key Constraints

3.) Entity Integrity Constraint

4.) Referential Integrity Constraint

5.) Not Null Constraint

6.) Unique Constraint

Design Issues

Design Issues are the problems or considerations that must be addressed while designing a relational database to ensure efficiency, consistency, & minimal redundancy.

Main Design Issues

1.) Data Redundancy

4.) Deletion Anomaly

2.) Update Anomaly

5.) Null values

7.) Dependency Preservation

3.) Insertion Anomaly

6.) Lossless Join

Summary :

Entity Set

An Entity Set is a collection of similar entities that share the same attributes.

Ex:- Entity: • Student 1 (Rahul)
• Student 2 (Priya)
• Student 3 (Aman)

— These are the Student Entity set.

Types of Entity Set :- 1.) Strong Entity Set - A strong Entity Set has its own Primary Key & can be uniquely identified without depending on any other entity.

- Has its own Primary Key
- Independent
- Single Rectangle
- Ex:- Student

2.) Weak Entity Set - A weak ~~entity~~ Entity Set does not have its own Primary Key & depends on a Strong Entity Set for identification.

Summary :

- Has no Primary Key
- Depends on Strong Entity
- Double Rectangle

Ex:- Dependent

Keys

A key is an attribute or a set of attributes that ~~uniquely identifies~~ uniquely identifies a record (tuple) in a table.

1.) Primary Key :- A Primary key is a key that uniquely identifies each record in a table. It cannot contain NULL or duplicate values. Ex:- Student-ID

2.) Candidate Key :- A candidate key is a minimal Super key that can uniquely identify each record. Ex:- Student-ID, Email. $\Rightarrow$ Both are candidate keys because both are unique.

3.) Super Key :- A Super key is a set of one or more attributes that uniquely identifies a record.

Ex:- {Student-ID} {Student-ID, Email}
 {Student-ID, Name}

Summary :



4.) Alternate Key :- An alternate key is a Candidate key that is not selected as the Primary Key.

Ex:- Primary Key = Student-ID

Alternate Key = Email

5.) Foreign Key :- A Foreign key is an attribute in one table that refers to the Primary key of another table.

Ex:- Student
Student-ID | Name

Course

Student-ID | Course

Course. Student-ID = Foreign Keys

6.) Composite Key :- A Composite key is a key made up of two or more attributes that together uniquely identify a record.

Ex:- Enrollment

Student-ID	Course-ID
101	C101
101	C102

Summary :



Composite key: (Student-ID, Course-ID)

⇒ Unique Key :- A Unique key ensures that all values in a column are unique. It does not allow duplicate values.

Ex:- Student

Student-ID	Email
101	abc@gmail.com
102	xyz@gmail.com

Unique key: Email

PCSAFCU → "Please Come Study After Finishing Computer University"

- * A Super Key is a superset of a Candidate key.
- * A Candidate key is a subset of Super key.
- * All primary keys are Super keys.
- * All Super keys are not Primary keys.
- * A Primary key cannot contain NULL values.
- * A Foreign key can contain NULL values.
- * A Table can have only one Primary key.
- * A Table can have multiple Candidate keys.

Summary :

Generalization

Generalization is the process of combining two or more lower-level entities that have common attributes into a higher-level entity.

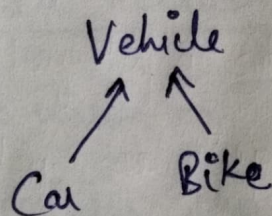
Ex:- Car (Vehicle No., Color)

Bike (Vehicle No., Color)

Both Car & Bike have common attributes, so they can be generalized into a higher-level entity vehicle.

⇒ Combining similar entities into a higher-level entity.

⊙ Generalization is a bottom-Up approach.

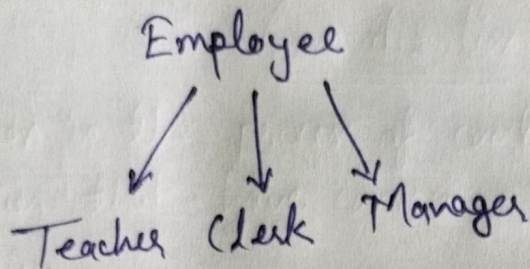


Specialization

Specialization is the opposite of generalization. It divides a higher-level entity into lower-level entities based on their specific characteristics.

Summary :

Ex:-



⇒ Specialization is a Top-down approach.

Dividing a higher level entity into lower-level entities.

Aggregation

Aggregation is a process in which a relationship set is treated as a higher-level entity so that it can participate in another relationship.

⇒ Aggregation is a Relationship Abstraction.

Ex:- 1) Employee works on Project
Manager monitors this work

Here, the relationship "Work on" is treated as a single entity so that it can participate in another relationship.

Aggregation is a concept used when a relationship itself needs to participate in another relationship.

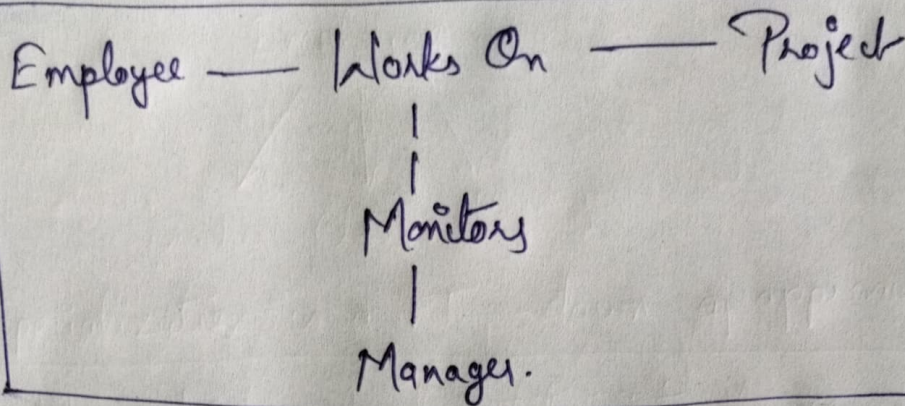
Summary :



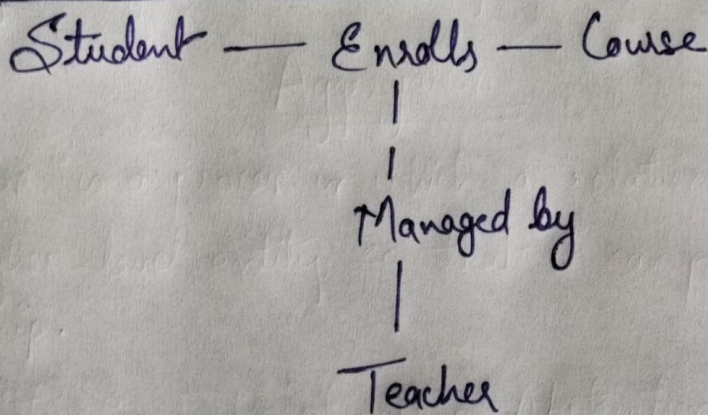
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Exo-2. →



Summary :